

FINAL EXAM

Fall 2020

- This is an **open**-book exam.
- No communication with other students is permitted. Any evidence of communication during the exam will be considered a violation of the SOM Honor Code.
- There are 180 points in the exam. You also have 180 minutes.
- Do not be concerned if your answers are not whole numbers.
- You will take the exam via Canvas; once you begin, you will have 3 hours.
- The exam will be in the form of a Canvas "Quiz". There is nothing to print.
- After the exam, you are welcome to scan and upload any scratch work for partial credit. There will be a [Dropbox link that you can click and then upload your scratch work](#). **Please label your work clearly if you would like partial credit.** Your file should be called lastname_firstname.pdf. You will have 20 minutes after the end of the exam to scan and upload your work.

Game Theory (45 Points)

Question 1: Price competition and capacity [30 points]

A-driving school and H-driving school are two competing driving schools in a town. Each can serve at most **80** students every week (and suppose each student takes a 1-hour training every week). The (weekly) cost of serving each student is \$10.

The two driving schools compete on price. Each can choose an hourly rate: \$50 or \$40. Suppose each driving school aims to maximize its OWN weekly profit, and the schools set their prices simultaneously.

The following is the **demand** matrix which shows how many students want to choose each driving school at each possible price combination (**before** taking into account the capacity constraint). For example, if A charges \$50 and H charges \$40, then 40 students want to choose A and 110 students want to choose H.

		H	
		\$50	\$40
A	\$50	60, 75	40, 110
	\$40	75, 65	70, 90

Note that, due to the capacity constraint, H cannot always serve its entire demand. For example, when it has demand 110, it cannot serve 30 students. **Half** of those students who choose H but cannot be served will go to A-driving school and pay A's price if A still has open seats. The other half will leave the market (for instance, because they do not like A-driving school and would rather to go to a driving school in the neighboring town).

Given the current capacities, draw the (weekly) profit matrix of the price competition. Don't forget the cost while computing profits! (You DON'T need to upload the matrix.) Then answer the following questions:

The effective DEMAND matrix, given H's current capacity constraint is:

		H	
		\$50	\$40
A	\$50	60, 75	55, 80
	\$40	75, 65	75, 80

Then the profit matrix is as follows:

		H	
		\$50	\$40
A	\$50	2400, 3000	2200, 2400
	\$40	2250, 2600	2250, 2400

(a): Does any driving school have a dominant strategy? [5 points]

H has a dominant strategy: \$50.

(b) What prices will the two schools charge in the Nash equilibrium? [5 points]

Each driving school will charge \$50.

Suppose H-driving school now expands its capacity to **110**. Suppose doing so involves a (weekly) fixed cost \$200, but it does not affect the marginal cost. Draw a new profit matrix by yourself. (Again you DON'T need to upload the graph.) Then answer the following questions:

If H-driving schools expands its capacity to 110, the effective demand matrix will be the one provided initially, and the new payoff matrix will therefore be as follows (after taking into account the fixed cost \$200:

		H	
		\$50	\$40
A	\$50	2400, $3000-200=2800$	1600, $3300-200=3100$
	\$40	2250, $2600-200=2400$	2100, $2700-200=2500$

(c) What prices will the two schools charge now? [5 points]

Now each school will charge \$40.

(d) How much (weekly) profit will H-driving school make? [5 points]

2500

The expansion will intensify price competition and drop its profit from 3000 to 2500. (Even without the expansion cost, its profit is only 2700, lower than before.)

(e) Suppose now that H-driving school also becomes a price leader after the capacity expansion, so that it sets its price first before A-driving school moves. In this case, how much (weekly) profit will H-driving school make? [5 points]

2800

In the sequential pricing game, backward induction predicts that H will charge \$50, and A will follow and charge the same price. So H-driving school will make a profit 3000, the same as before expansion.

(f) Instead of expanding its capacity, H-driving school can alternatively sign a contract with A-driving school to buy its unused capacity. By paying \$15 for each student, H-driving school can send its unserved students to A-driving school (up to its capacity). For each of those students, A-driving school will then bear the cost \$10. (Suppose A-driving school is always willing to accept this contract, and they set prices simultaneously afterwards.) Facing the three options: retain the status quo, expand capacity, or buy A's unused capacity, which option will H-driving school choose? [5 points]

Remain the status quo.

If H-driving school signs that contract with A-driving school, the new payoff matrix in the price competition game will be:

		H	
		\$50	\$40
A	\$50	2400, 3000	1750, 3150
	\$40	2250, 2600	2150, 2650

[The payoff numbers in the two cells on the left are the same as before because the capacity constraints do not bind. The payoff numbers in the two cells on the right can be explained as follows: For example, consider the cell when A charges \$50 and H charges \$40. In that case H will send 30 students to A. So H's profit is $80 \cdot 30 + 30 \cdot 25 = 3150$, because it makes a profit \$30 from each of the 80 students served by itself and a profit \$25 from each of the 30 students sent to A. A's profit is $40 \cdot 40 + 30 \cdot 5 = 1750$, because it makes a profit \$40 from each of its own students and a profit \$5 from each of the students from H. Similarly, the other cell on the right can be explained.]

From the payoff matrix, we can see that H has a dominant strategy \$40, and anticipating that A will then choose \$40 as well. So H will end up making a profit \$2,650, which is still lower than the initial profit 3000.

Question 2: A game of switching envelopes [15 points]

You and your friend are playing a game of switching envelopes. There are two envelopes, each containing an amount of money; the amount of money is either \$2, \$4, \$8, \$16, \$32, or \$64, and both of you know this. Furthermore, both of you are told that one envelope contains exactly twice as much money as the other. The two envelopes are shuffled, and one is given to you and the other is given to your friend. Both of you open your envelope secretly (so that the amounts inside are kept private), and then are given an opportunity to switch. If both of you agree to switch, you switch your envelopes; if at least one of you doesn't want to switch, you keep your envelopes.

(a) If you see \$16 in your envelope, should you offer to switch? [5 points]

No, you shouldn't offer to switch.

This can be seen by a reasoning similar to backward induction. If a player receives an envelope with \$64, she will never want to switch. Knowing that, a player who receives \$32 should not offer to switch for the only time the other player might trade with her occurs when the other player has \$16. Knowing that, a player who receives \$16 shouldn't offer to switch either.

(b) If you see \$4 in your envelope, should you offer to switch? [5 points]

No. The same reasoning.

(c) Suppose the above underlined sentence is replaced by "both of you are told that the two envelopes contain different amounts of money". If you see \$8 in your envelope, should you offer to switch? [5 points]

No. That sentence doesn't affect the reasoning qualitatively. A similar argument as before applies.

Section II: Short Questions [60 points]

Please be concise. None of the answers requires more than a few sentences.

- 1) [10 points] The Indian government is contemplating the imposition of a price ceiling on an existing anti-viral drug that is proving to be effective in treating some of the symptoms of COVID-19. The patent for this drug has expired and the market for this drug is perfectly competitive, but the market is not at its long-run equilibrium since the COVID-19 situation is still evolving. Patient groups that typically advocate in favor of patient rights are vehemently opposed to this price ceiling. What is an economic rationale for their argument?

With the price ceiling, supply will fall and some patients will not be able to buy the drug who had been able to buy it before.

On the whole, patients will lose if supply is pretty elastic; the decrease in consumer surplus from the fall in Q outweighs the increase in CS from those who still buy the drug.

- 2) [10 point] You are manufacturing cloth facemasks for your cohort at SOM. There are two designs: One is a fancy 3 ply mask with adjustable ear loops with a visual of the bulldog (the B-mask) and the other is a single-ply simple design in blue fabric with the Yale University Y (the Y-mask). You make these masks on a state-of-the-art sewing machine that you can use for free at the Yale Fabric Arts Studio. In one hour on the machine, you can produce one fancy B-masks using a single worker who earns a wage of \$15 per hour. To make a Y-mask you require only thirty minutes on the machine, but for those thirty minutes you need two workers (at the same rate of \$15 per hour) are required. For either type, a mask uses \$5 worth of material. Since manufacturing Y or B masks takes \$15 worth of labor, uses \$5 worth of material and is done on a machine you can use for free, you are thinking of pricing them the same. Assuming you simply want to maximize profit, is this a good decision? Explain why, or why not. Provide some economic intuition, using numbers.

You cannot forget the opportunity cost of the machine. For every hour of machine time spent on making Y-masks, you earn $2(p_Y - \$20)$. For every hour spent on B-masks you make $(p_B - 20)$.

Simplifying, if $2p_Y - p_B > 20$ you should make only Y-masks; if the inequality goes the other way make B-masks and with the equality holding, the firm will be indifferent on product mix.

- 3) [10 points] Oregon is considering imposing a high specific excise tax on manufacturers of cigarettes, cigars and vaping products. Voters will weigh in on this measure in November. Opponents of the tax make two arguments: First they claim that the proposed tax of \$3.33 per package is too high and will drive companies out of business as they cannot afford to pay \$3.33 out of the profit for every pack sold. Second, they argue that taxes are distortionary, and in this tax will create a lot of deadweight loss. What economic reasoning can you offer to counter the above two specific arguments?

Incidence of tax shared between companies and consumers, and since cigarette demand is inelastic, burden of tax will be on consumers.

Taxes may be distortionary but since demand is inelastic, the extent of distortion is small. Or there are negative externalities – and so measuring deadweight loss just in the cigarette market is not a reasonable way to measure the impact of this tax.

- 4) [10 points] You are in the market to buy a used car. Some sellers offer warranties and others don't. You choose to go with a seller who offers a service warranty for two years, but with a \$500 deductible.
- Does offering a warranty address the problem of adverse selection, moral hazard, neither, or both?
 - Similarly, does the clause about the \$500 deductible address the problem of adverse selection, moral hazard, neither, or both? Explain your reasoning clearly
 - Your classmate from SOM tells you about Carfax which is a website that provides information about whether a used car is low-quality or high-quality and what it is worth. Do you think such a service improves how the market functions? Under what conditions can such a service enable the used car market achieve the first-best outcome?

Warranty: Addresses the adverse selection problem. Costly signal

Deductible: Addresses moral hazard. Helps prevent risk taking behavior by driver

Yes, can improve. But first best only if services are free.

- 5) [10 points] The market for Hawaiian Kona coffee is perfectly competitive. Your analysis of the financial statements of a large coffee producer in the region reveals that the firm has been making millions of dollars of profits consistently for the last ten years. Is this consistent with your understanding that long-run economic profits of a firm in a perfectly competitive market should be zero? Provide a precise explanation for your answer.

One of the following explanations

(1) Economic profits can be zero while accounting profits can still be positive. Financial statements show accounting profits – so ignores opportunity costs.

(2) When firms have economies of scale (e.g., when the ATC is U shaped), in the short run firms can earn a positive profit.

(3) Alternatively, marginal firms in a competitive market makes zero economic profit, but this firm is more efficient than the marginal ones.

- 6) [10 points] The Government of Japan imposes a very high import duty on copper products from Chile, Peru and Australia. Who do you think would be the gainers and losers in Japan from such legislation?

Producers of copper products in Japan would gain

Consumers would pay higher prices and so would lose

Government would get tax revenue

Section III: Long Questions [75 points]

Problem 1 (15 points)

Google has introduced a new product known as a “Jamboard”, which is an interactive “smart” whiteboard. They have identified two distinct market segments for this product: Higher Ed (“H”), who can use this for teaching, and VC-Fueled Startups (“V”), who have more funding than they know what to do with. These segments have shown a lot of interest now that many workers are working remotely and many universities are doing distance learning. Fall demand for Jamboards for each subgroup is given by:

$$Q_H = 2000 - P_H$$

$$Q_V = 4000 - 0.5P_V$$

Google estimates it has a constant marginal cost of \$500 per Jamboard.

- a) (4 points) What is the profit-maximizing price for a Jamboard to each group?

Price to Higher Ed:

Price to VC-Fueled Startups:

Answer: \$1250 to H, \$4250 to V. Two points for each, one point if set up correctly but a math error. This comes from setting $MR=MC$ for each of these (invert to get $P=$ demand, multiply by Q to get revenue, take the derivative to get MR , set equal to 500).

- b) (6 points) Google would be able to price discriminate in this way if it can verify who is buying a Jamboard and ask Higher Ed customers to prove their status. However, some clever startups may claim to be a Higher Ed institution to get a better price. What price would Google charge for a Jamboard if they could not verify a buyer’s type, e.g. if any startup could claim to be an institution of Higher Ed?

Price for Jamboard:

Answer: If you look at the combined demand curve, the answer looks like \$2,250, HOWEVER, at that price, no H are buying. SO, the answer is 4250 (the optimal price for V). This is a bit tricky, since you need to check which part of the demand curve you are on after solving for the price of \$2,250. You can verify by comparing charging a price of \$2250 to only the V segment; that is clearly lower than charging \$4250.

- c) (5 points) An accounting firm has approached Google and offered to audit buyers who claim to be Higher Ed to confirm their true status, but they have asked for \$1M for their services for the fall. Would Google be willing to pay this, and why?

Answer: No, because profits don’t increase enough from the price discrimination.

Problem 2 (15 points)

The COVID situation had led to much uncertainty in the travel industry. You are thinking about booking a trip over spring break in 2021, and the vacation package you are considering is to a Caribbean island and costs \$3,000. You have \$20,000 in savings and your utility function for your wealth is $u(x) = \sqrt{x}$ where x is measured in thousands, so that your utility of \$20,000 is $\sqrt{20}$.

The vacation package is non-refundable. However, you can purchase travel insurance for \$750 that will refund your purchase if the US issues a travel advisory to this particular island covering spring break. You are not willing to violate a travel advisory, so if you purchase the vacation package without buying insurance and an advisory is issued, you will lose your \$3,000.

You estimate that there is a 20% chance of the US issuing a travel advisory that would prevent you from traveling over spring break. To be clear, you value the vacation at \$3,000, so if you get to take the vacation, it is as if your wealth is \$20,000, but if you pay for it and don't get to take it, you lose the \$3,000.

1. (4 points) Will you choose to buy this insurance?

Answer: EU of taking risk: $EU = 0.2 * \sqrt{20 - 3} + 0.8 * \sqrt{20} = 4.402$
 $EU_{insure} = 1 * \sqrt{19.25} = 4.387$

No, you will not buy this insurance.

2. (6 points) Is this insurance priced in an actuarially fair way (assuming the insurance company agrees with your 20% probability)? Would a person ever purchase insurance if it is not priced in an actuarially fair way?

Answer: No. Expected cost is $0.2 * 3000 = \$600$, but price is \$750
 BUT, people would buy if they are sufficiently risk averse. Full credit if they mention gains from trade between risk averse individuals and risk neutral firm; or, if they simply say that someone who is sufficiently risk averse would buy it.

3. (5 points) What is the most you would be willing to pay for this insurance policy?

Answer: There are a few ways to determine this. I will compute the certainty equivalent.

$$\sqrt{c} = 0.2 * \sqrt{20 - 3} + 0.8 * \sqrt{20}$$

$$c = 19.3805$$

Therefore, the most I would pay is \$619 (+/- 10 for rounding).

Problem 3 (15 points)

2. (15 points) The producers of the upcoming film, "Star Wars Episode X: The Search for More Money", have been debating how to release the film for maximum profit. Disney knows that there are two types of fans: "Knights", who are very eager to see the film, and "Padawans", who are

interested but not avid followers of the franchise. The producers have done market testing and discovered the following willingness-to-pay for both a regular and premium (3D with souvenirs) theatrical version of the film for each of the two segments. Assume everyone wants to go just one time and that there is not a global pandemic occurring. They have also determined that there are **four Padawans for every one Knight**.

Reservation Prices:	Knights (1)	Padawans (4)
Regular Film (MC = 0)	20	15
3D Film (MC = 8)	45	25

From the producers' point of view, the marginal cost of the regular format film is zero, while for the 3D version, it is \$8 to cover licensing, 3D glasses, and souvenirs.

- a. (4 points) Suppose the producers only wanted to target the "Knights" fans. What is the optimal price and which version of the film should they release?

☐ Release Regular Film

☐ Release 3D Film

Optimal Price: _____

SOLUTION: Release 3D at a price of \$45. Profit of $$(45-8)=37$ per is greater than if we price the regular at \$20.

- b. (4 points) Now consider that the producers want to charge a price that attracts **both segments**, still only releasing **one version** (Regular or 3D). What is the best price?

☐ Release Regular Film at a price of: _____

OR

☐ Release 3D Film at a price of: _____

SOLUTION: You're charging the low types' reservation price in either case, but the margin is higher on the 3D film, so release 3D at a price of 25.

- c. (4 points) The producers are now considering offering **both formats** of the film at prices such that fans self-select into the different formats. What prices should they charge to maximize profits under this scheme?

Release Regular Film at a price of: _____

Release 3D Film at a price of: _____

**SOLUTION: First, make sure the low types are willing to buy: release regular film at \$15;
Then, make sure high types buy high good. That means release 3D at \$40 (or \$39.99).**

- d. (3 point) Which scheme should they follow in order to maximize profits?

☐ Only release the Regular film

☐ Only release the 3D film

☐ The self-selecting scheme

SOLUTION: Self-selecting yields profit of 15 for each low type and 32 for each high type. Since there are 4:1 low to high types, that's \$92 in profit relative to $$(17 \cdot 5) = 85$ when releasing 3D at 25. So, self-selecting is best.

(15 points) Problem 4: Demand for outdoor dining in New Haven is at an all-time high. The many restaurants located downtown are finding new outdoor spaces and diners very much appreciate the ability to go to a restaurant but be seated outdoors where the risk of virus transmission appears to be lower.

To be specific, the number of outdoor meals that downtown New Haven restaurants can supply in a week is given by

$$Q_S = 250P - 200$$

Where P is the price of a meal. Demand for outdoor meals is given by

$$Q_D = 6100 - 50P$$

1. (4 points) What is the equilibrium price and quantity for outdoor dining meals in New Haven?

Answer: $250P - 200 = 6100 - 50P$

$$6300 = 300P$$

$$P = 21$$

$$Q = 5050$$

2. (3 points) At this equilibrium, what is the elasticity of demand for a meal in New Haven? (Please answer to 2 digits after the decimal)

Answer: $\epsilon = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q} = -50 \cdot \frac{21}{5050} = -0.208, \text{ +/- } 0.01$

3. (5 points) The city recognizes that there is a negative externality from individuals going out to eat, versus staying home. However, the city is also concerned about tax revenues falling due to lower economic activity in other sectors. Therefore, they are thinking of imposing a tax of \$1 per outdoor meal to raise new revenue. If they imposed such a tax, what would the new equilibrium price and quantity be?

Answer:

$$P_S = \frac{1}{250}Q + 0.8 + 1 = \frac{1}{250}Q + 1.8$$

$$Q_S = 250P - 450 = 6100 - 50P = Q_D$$

$$300P = 6550$$

$$P = 21.83$$

$$Q_D = 5008$$

4. (3 points) The city is hoping that the tax raises at least \$4,000 per week to help provide relief for those suffering from the effects of the pandemic. How much tax revenue would this raise per week?

Answer: It raises ~\$5,008 per week (would accept +/- 50).

Problem 5

(15 points) A local clothing retailer currently pays its sales staff a fixed hourly rate of \$15 per hour, which is the prevailing wage for sales staff in this line of work at various clothing stores in New Haven. The owner is considering switching to a wage and bonus structure to better incentive her sales staff. She spent some time carefully watching her staff, collecting data, and found the following: on average, five customers come by per hour (with one salesperson working per hour). If a salesperson is attentive and helpful for an hour, each of the 5 customers has a 40% chance of making a purchase, while if the sales staff are otherwise distracted, the chance of each customer making a purchase is 20%. Furthermore, based on her own background and experience, she feels that for a salesperson, the “cost of effort” of being attentive and helpful is around \$6 per hour. (It is costless to be distracted.) Assume that salespeople are risk-neutral.

- a) (3 points) What is the “first best” (perfect information) wage that should be offered here? Suppose there were a fixed wage structure with perfect monitoring. The optimal hourly wage from the view of the firm, where any distracted employees would be fired is:

Answer: outside option + cost of effort = 21

- b) (8 points) Now consider the reality of poor monitoring. What is the optimal wage and bonus structure – the lowest possible wage and bonus (**per customer who makes a purchase**) such that her scheme will result in her sales staff exerting more effort -- to offer her sales staff?

Answer:

Willing to work constraint

$$w + 5 * .4 * b - 6 = 15$$

$$w = 21 - 2b$$

Do the right thing

$$w + 2b - 6 = w + 5 * 0.2 * b$$

$$b = 6$$

So, wage of 9 and bonus of 6 per customer that buys

- c) (4 points) Clothing is a tough market, but the owner estimates she makes a profit margin of \$10 per customer that makes a purchase. Is the wage-and-bonus scheme worthwhile for her? Or should she stick with the fixed \$15 per hour salary structure (with the understanding that sales staff would not be nearly as attentive and helpful under that approach)? Explain your reasoning.

Answer: yes to the wage-and-bonus. A few ways to reach this conclusion

- “the gain to owner should be more than the cost of bonus/effort”: the effort increases the expected hourly profits from $5 * .2 * 10 = 10$ to $5 * .4 * 10 = 20$, but also means paying out \$15 vs $\$9 + 5 * .4 * 6 = 21$ in wages.
- So I am “spending” \$6 more to make \$10. Good deal.
- $-15 + 1 * 10 = -5$ is the EV of an hour of time under fixed wage
- $-9 + 5 * .4 * 10 - 5 * .4 * 6 = -1$ is the EV of an hour of time under the wage/bonus